GSE 518 MIDTERM CHEAT SHEET • SIDE 1: THE MACHINES

0. SYMBOL DECODER

X random variable (the machine) | x a value it takes | $\mathcal{X}$ the **support**, where $f_X > 0$ | $\mu = \mathbb{E}[X]$ center | $\sigma^2 = \mathrm{Var}(X)$ spread^2 | $\sigma = \sqrt{\sigma^2}$ | ρ correlation, in $[-1,1]$ | Φ standard normal *cdf*, ϕ its *pdf* | $g^{-1}(y)$: solve $y = g(x)$ for x. **In $N(\mu, \sigma^2)$ the second slot is the VARIANCE. Standardize by σ.**

1. CDF / PMF / PDF

$F_X(x) = \mathrm{P}(X \le x)$. $f_X(x) = \frac{\mathrm{d}F_X}{\mathrm{d}x}$, $F_X(x) = \int_{-\infty}^{x} f_X(r)\,\mathrm{d}r$.
Valid pdf/pmf: (i) $f \ge 0$; (ii) integrates/sums to 1.
Valid cdf: (i) limits 0 and 1; (ii) non-decreasing; (iii) right continuous.
Discrete: $f_X(x) = \mathrm{P}(X = x)$, jump height of F. Continuous: f is NOT a probability, can exceed 1; $\mathrm{P}(X = x) = 0$.
Write cdfs piecewise on all of $\mathbb{R}$, including the 0 and 1 branches.

2. EXPECTATION AND VARIANCE

$$\mathbb{E}[g(X)] = \textstyle\sum_x g(x) f_X(x) \ \text{ or } \ \int_{\mathcal{X}} g(x) f_X(x)\,\mathrm{d}x$$

$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$; $\mathbb{E}[c] = c$; $\mathbb{E}[g(X)] \ne g(\mathbb{E}[X])$.

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2} \qquad \mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

If the integral diverges, the moment **does not exist**. Say so.

3. TRANSFORMATIONS $Y = g(X)$ (4 STEPS)

(1) Find $\mathcal{Y}$ by pushing endpoints of $\mathcal{X}$ through g. **(2)** Check g strictly monotone *on* $\mathcal{X}$ via sign of $\mathrm{d}g/\,\mathrm{d}x$. **(3)** Invert: $x = g^{-1}(y)$; differentiate. **(4)** Apply:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{\mathrm{d}g^{-1}(y)}{\mathrm{d}y} \right| \text{ on } \mathcal{Y},\ 0 \text{ o/w}}$$

DISCRETE: same but NO Jacobian: $f_Y(y) = f_X(g^{-1}(y))$.
cdf route: $g \uparrow$: $F_Y(y) = F_X(g^{-1}(y))$; $g \downarrow$: $F_Y(y) = 1 - F_X(g^{-1}(y))$.
Never forget: absolute value + state the support.

4. NORMAL SHORTCUTS

$$X \sim N(\mu, \sigma^2),\ Y = aX + b \ \Rightarrow\ \boxed{Y \sim N(a\mu + b,\ a^2\sigma^2)}$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0,1); \quad \mathrm{P}(X \le c) = \Phi\left(\frac{c - \mu}{\sigma}\right)$$

$\Phi(-z) = 1 - \Phi(z)$; $\phi(-z) = \phi(z)$; $\Phi(0) = 0.5$; $Z^2 \sim \chi^2_{(1)}$.
No closed form for the normal cdf; use the table.

5. JOINT DISTRIBUTIONS (HALF THE EXAM)

Marginal: $f_X(x) = \int_{\mathcal{Y}} f_{XY}(x,y)\,\mathrm{d}y$ (integrate out the OTHER)

Conditional: $f_{Y|X}(y|x) = \dfrac{f_{XY}(x,y)}{f_X(x)}$ (joint ÷ marginal of the GIVEN one)

Cond. exp.: $\mathbb{E}[Y|X = x] = \int_{\mathcal{Y}} y\, f_{Y|X}(y|x)\,\mathrm{d}y$ (a function of x)

$$f_{XY} = \frac{\partial^2 F_{XY}}{\partial x \partial y}; \quad \mathbb{E}[g(X,Y)] = \iint g(x,y) f_{XY}\,\mathrm{d}x\,\mathrm{d}y$$

Range belongs to the variable LEFT of the bar.
Ask every time: which variable is random? Integrate over that one.

6. INDEPENDENCE

Factorization Thm (Lemma 4.2.7): $X \perp\!\!\!\perp Y \iff f_{XY}(x,y) = g(x)h(y)$.
Support must be a RECTANGLE. If limits like $0 < x < y < 1$, NOT independent even if it looks separable.
Sum $(x + y)$ does not factor. Product (xy) does. $e^{-(x+y)} = e^{-x}e^{-y}$ DOES.
If independent: $f_{Y|X} = f_Y$; $\mathbb{E}[r(X)s(Y)] = \mathbb{E}[r(X)]\mathbb{E}[s(Y)]$; $\mathrm{Cov} = 0$; $M_{X+Y} = M_X M_Y$; $g(X) \perp\!\!\!\perp h(Y)$.
$\mathrm{Cov} = 0$ **does NOT imply independence** (only linear dependence is measured). **Exception: bivariate normal, where $\rho = 0 \iff$ independent.**

7. COVARIANCE AND CORRELATION

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]; \qquad \rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; $\mathrm{Cov}(X,a) = 0$; Cov is symmetric.

$$\boxed{\mathrm{Cov}(aX + bY, cU + dV) = ac\mathrm{Cov}(X,U) + ad\mathrm{Cov}(X,V)}$$

$$\boxed{+bc\mathrm{Cov}(Y,U) + bd\mathrm{Cov}(Y,V)}$$

$$\boxed{\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\mathrm{Cov}(X,Y)}$$

Independent $\Rightarrow$ drop the cross term. **Expand like FOIL.**
$\rho = \pm 1 \iff Y = aX + b$ exactly (+ if $a > 0$). ρ is unitless. **If $|\rho| > 1$ you made an error.**

8. LIE AND JENSEN

$$\boxed{\mathbb{E}[Y] = \mathbb{E}_X[\mathbb{E}[Y|X]]} \qquad \mathbb{E}[r(Y)] = \mathbb{E}_X[\mathbb{E}[r(Y)|X]]$$

$$\mathbb{E}[Y|X] = \mathbb{E}_{Z|X}[\mathbb{E}[Y|X,Z] \mid X]$$

Outer expectation uses the marginal of whatever was RIGHT of the bar.
Jensen: g convex $\Rightarrow \mathbb{E}[g(X)] \ge g(\mathbb{E}[X])$; concave $\Rightarrow \le$.
$\mathbb{E}[X^2] > (\mathbb{E}[X])^2$; $\mathbb{E}[\ln X] < \ln \mathbb{E}[X]$; $\mathbb{E}[1/X] > 1/\mathbb{E}[X]$.

9. BIVARIATE NORMAL $N(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$

Marginals normal. $\rho = 0 \iff$ independent.

$$Y|X \sim N\left(\mu_Y + \rho\tfrac{\sigma_Y}{\sigma_X}(X - \mu_X),\ \sigma_Y^2(1 - \rho^2)\right)$$

$\mathbb{E}[Y|X] = a + bX$ is LINEAR; slope $b = \rho\frac{\sigma_Y}{\sigma_X} = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$ **= OLS β.**

$$aX + bY \sim N\left(a\mu_X + b\mu_Y,\ a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y\right)$$

10. PROBABILITY AND BAYES

$\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$; $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$;
$A \subseteq B \Rightarrow \mathrm{P}(A) \le \mathrm{P}(B)$; DeMorgan $(A \cup B)^c = A^c \cap B^c$.
Boole: $\mathrm{P}(\cup A_i) \le \sum \mathrm{P}(A_i)$. Bonferroni: $\mathrm{P}(\cap A_i) \ge 1 - \sum[1 - \mathrm{P}(A_i)]$.

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}; \quad \mathrm{P}(A \cap B) = \mathrm{P}(B)\mathrm{P}(A|B)$$

Total prob: $\mathrm{P}(B) = \sum_i \mathrm{P}(B|A_i)\mathrm{P}(A_i)$

$$\boxed{\textbf{Bayes: } \mathrm{P}(A_i|B) = \frac{\mathrm{P}(B|A_i)\mathrm{P}(A_i)}{\sum_j \mathrm{P}(B|A_j)\mathrm{P}(A_j)}}$$

$\mathrm{P}(A|B) + \mathrm{P}(A^c|B) = 1$, but $\mathrm{P}(A|B) + \mathrm{P}(A|B^c) \ne 1$.
Independent events: $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$. For 3+, need ALL subsets; pairwise $\ne$ mutual.

11. INEQUALITIES

Markov ($X \ge 0, c > 0$): $\mathrm{P}(X \ge c) \le \mathbb{E}[X]/c$.
Chebyshev (any $X, k > 0$): $\mathrm{P}(|X - \mu| \ge k) \le \sigma^2/k^2$.
With $k = m\sigma$: $\mathrm{P}(|X - \mu| \ge m\sigma) \le 1/m^2$.

12. DISTRIBUTIONS: $\mathbb{E}[X]$, $\mathrm{Var}(X)$, mgf

Dist	$\mathbb{E}[X]$	$\mathrm{Var}(X)$	$M_X(t)$
Bernoulli(p)	p	$p(1-p)$	$(1-p)+pe^t$
Binomial(n,p)	np	$np(1-p)$	$[(1-p)+pe^t]^n$
Geometric(p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	
NegBin(r,p)	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	
Poisson(λ)	λ	λ	$e^{\lambda(e^t-1)}$
Disc.Unif$\{1..N\}$	$\frac{N+1}{2}$	$\frac{N^2-1}{12}$	
Uniform(a,b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
Normal(μ,σ^2)	μ	σ^2	$e^{\mu t+\sigma^2t^2/2}$
Gamma(α,β)	$\alpha\beta$	$\alpha\beta^2$	$(1-\beta t)^{-\alpha}$
Expon(λ)	λ	λ^2	$(1-\lambda t)^{-1}$
$\chi^2_{(p)}$	p	$2p$	$(1-2t)^{-p/2}$
Beta(α,β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	
$t_{(r)}$	0 $(r>1)$	$\frac{r}{r-2}$ $(r>2)$	none

pdfs: Bern $p^x(1-p)^{1-x}$ | Bin $\binom{n}{x}p^x(1-p)^{n-x}$ | Geom $p(1-p)^{x-1}$, $x\geq 1$ | Poisson $\frac{e^{-\lambda}\lambda^x}{x!}$ | Unif $\frac{1}{b-a}$ | Normal $\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/2\sigma^2}$ | Gamma $\frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}$ | $\chi^2_{(p)}$ $\frac{x^{p/2-1}e^{-x/2}}{\Gamma(p/2)2^{p/2}}$ | Beta $\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$.

Prof's exponential convention: $f=\frac{1}{\lambda}e^{-x/\lambda}$, $\mathbb{E}=\lambda$. If given ce^{-cx}, call it "exponential with mean $1/c$".

Identities: Gamma$(1,\beta)$ =Exp(β); Gamma$(\frac{p}{2},2)$ = $\chi^2_{(p)}$; Beta$(1,1)$ =Unif$(0,1)$; Bin$(1,p)$ =Bern(p); NB$(1,p)$ =Geom(p); $Z^2\sim\chi^2_{(1)}$; $\sum\chi^2_{(p_i)}\sim\chi^2_{(\sum p_i)}$; $F(1,r)=t^2_{(r)}$.

13. MGF TOOLKIT

$$M_X(t)=\mathbb{E}[e^{tX}];\qquad \mathbb{E}[X^n]=M_X^{(n)}(0)$$

Thm 2.3.11: $M_X=M_Y$ near $0\Rightarrow X\sim Y$. **Cite it by name.**
Thm 4.2.12: $X\perp\!\!\!\perp Y\Rightarrow M_{X+Y}=M_X\cdot M_Y$.
Uniform mgf undefined at $t=0$, so it cannot generate moments.
Trick: to kill a hard integral, reshape the integrand into a known pdf and set the integral to 1.

14. $\Phi(z)$ TABLE (standard normal cdf)

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.00	.5000	0.90	.8159	1.80	.9641
0.10	.5398	1.00	.8413	1.90	.9713
0.25	.5987	1.10	.8643	1.96	.9750
0.50	.6915	1.20	.8849	2.00	.9772
0.60	.7257	1.28	.8997	2.33	.9901
0.675	.7500	1.40	.9192	2.50	.9938
0.75	.7734	1.645	.9500	2.58	.9951
0.80	.7881	1.75	.9599	3.00	.9987

$\Phi(-z)=1-\Phi(z)$. Central intervals: $\pm1\sigma\approx 68\%$, $\pm1.645\sigma=90\%$, $\pm1.96\sigma=95\%$, $\pm2.58\sigma=99\%$.

15. CALCULUS AND SERIES FACTS

$$\int x^n\,dx=\tfrac{x^{n+1}}{n+1};\quad \int\tfrac{1}{x}\,dx=\ln x;\quad \int e^{ax}\,dx=\tfrac{e^{ax}}{a}$$

$$\int_0^\infty e^{-ax}\,dx=\tfrac{1}{a};\quad \int_0^\infty xe^{-ax}\,dx=\tfrac{1}{a^2};\quad \int_0^\infty u^{\alpha-1}e^{-u}\,du=\Gamma(\alpha)$$

$$\Gamma(1)=1,\Gamma(\tfrac{1}{2})=\sqrt{\pi},\Gamma(\alpha+1)=\alpha\Gamma(\alpha),\Gamma(n)=(n-1)!$$

$$\sum_{i=1}^n i=\tfrac{n(n+1)}{2};\quad \sum_{i=1}^n i^2=\tfrac{n(n+1)(2n+1)}{6};\quad \sum_{i=0}^\infty ar^i=\tfrac{a}{1-r}\ (|r|<1)$$

$\binom{n}{x}=\frac{n!}{(n-x)!\,x!}$. Double integral: do the INNER variable first, treating the outer as constant.

16. THE PLAYBOOK: SIGNAL → MOVE

Sees	Does
"find c so it's a pdf"	set $\int f=1$, solve
"find the cdf"	$\int_{-\infty}^x f$; piecewise, all of $\mathbb{R}$
"$Y=g(X)$, find f_Y"	4-step change of variable
"$X\sim N$, $Y=aX+b$"	$N(a\mu+b,a^2\sigma^2)$
"$P(X\leq c)$", normal	standardize, $\Phi\left(\frac{c-\mu}{\sigma}\right)$
"find $\mathrm{Var}(X)$"	$\mathbb{E}[X^2]-(\mathbb{E}[X])^2$
"find the marginal"	integrate out the other one
"independent?"	Factorization + rectangle check
"find $f_{Y\|X}$"	joint ÷ marginal of X
"find $\mathbb{E}[Y\|X=x]$"	$\int yf_{Y\|X}\,dy$, over y
"$\mathbb{E}[\mathbb{E}[Y\|X]]$"	LIE $=\mathbb{E}[Y]$, do not integrate
"Corr(W,U), linear combos"	FOIL the Cov, then $\mathrm{Var}(aX+bY)$
"show $X\sim$ some dist"	mgf, match, cite Thm 2.3.11
"bound a probability"	Markov $(X\geq 0)$ or Chebyshev
"$\mathbb{E}[g(X)]$ vs $g(\mathbb{E}[X])$"	Jensen
"given $P(B\|A)$, want $P(A\|B)$"	Bayes
"a partition, then event B"	Law of Total Probability

17. DEDUCTION-KILLERS (check before submitting)

1. State the support on every density. **2.** cdf needs 0 and 1 branches. **3.** Absolute value in the Jacobian. **4.** Conditional's range belongs to the LEFT-of-bar variable. **5.** Standardize by σ, not σ^2. **6.** Marginal of X integrates out y. **7.** Never say uncorrelated $\Rightarrow$ independent (unless bivariate normal). **8.** NAME the theorem. **9.** If it says "use a result from class", do not integrate. **10.** Discrete: $P(X\geq 2)=1-P(X\leq 1)$. **11.** Check densities integrate to 1. **12. Show all work: setup earns more than the answer.**

18. THE PS6#2 / SM#5 TEMPLATE (do this in order)

Given f_{XY} on a rectangle: **(1)** write the support. **(2)** check $\iint f=1$. **(3)** $f_X=\int_{\mathcal{Y}}f\,dy$, $f_Y=\int_{\mathcal{X}}f\,dx$. **(4)** independence via factorization. **(5)** $f_{Y|X}=f_{XY}/f_X$, state y's range. **(6)** $\mathbb{E}[Y|X=x]=\int yf_{Y|X}\,dy$. **(7)** conditional probability: plug in the given value, integrate. **(8)** $\mathbb{E}[\mathbb{E}[Y|X]]=\mathbb{E}[Y]$ by LIE. **(9)** for ρ: need $\mathbb{E}X$, $\mathbb{E}Y$, $\mathbb{E}X^2$, $\mathbb{E}Y^2$, $\mathbb{E}[XY]=\iint xyf$; then $\mathrm{Cov}=\mathbb{E}[XY]-\mathbb{E}X\mathbb{E}Y$ and divide by $\sigma_X\sigma_Y$.

Known answers: PS6#2 $(0.25(2x+y),\ 0<x<1,\ 0<y<2)$: $f_X=x+\frac{1}{2}$, $f_Y=\frac{1+y}{4}$, $\mathbb{E}X=\frac{7}{12}$, $\mathbb{E}[X|Y=y]=\frac{1}{1+y}(\frac{2}{3}+\frac{y}{2})$. SM#5 ($x+y$ on unit square): $f_X=x+\frac{1}{2}$, $\mathbb{E}X=\mathbb{E}Y=\frac{7}{12}$, $\mathrm{Var}=\frac{11}{144}$, $\mathbb{E}[XY]=\frac{1}{3}$, $\mathrm{Cov}=-\frac{1}{144}$, $\rho=-\frac{1}{11}$, $\mathbb{E}[Y|X]=\frac{3x+2}{6x+3}$.